We will get to reading schematics shortly, but even so, hopefully, these two schematics will illustrate the differences of the NPN and PNP transistors. We use the NPN transistor, like in the schematic on the left, to switch a load on the low side or negative side of the load. The PNP, on the other hand, like in the schematic on the right, is used to switch the high side or positive side of the load.

Another kind of transistor is the Metal Oxide Field Effect Transistor or MOSFET. MOSFETs, such as the IRF540, are available in N-channel and P-channel varieties, similar to NPN and PNP transistors. Unlike transistors though, MOSFETs are capable of handling larger currents and switch faster than our run-of-the-mill transistors. They are best used for large motors, high-wattage LEDs, heaters, or similar applications.

Switches

Switches are the human-operable, mechanical version of transistors. Their job is to interrupt or allow current to flow through a circuit. They do this with metal contacts inside the switch that either close or open when the switch is activated. There are many different kinds of mechanical switches, including momentary pushbuttons, slide switches, lever switches, reed switches, tilt switches, toggle switches, and even relays are a kind of mechanical switch. Many of these are shown in Figure 12-8.



Figure 12-8. *Switches*

The contacts of a switch are either normally open, N.O., or normally closed, N.C. So for example, a circuit with a normally open momentary pushbutton with two contacts will be open until the button is pressed closing the circuit. When the button is released again, the circuit is once again open. A lever switch with three contacts might have one pin labeled N.O., another N.C., and a third C. for common. When a circuit is connected to the N.O. and C. contacts, the circuit will be normally open and closed